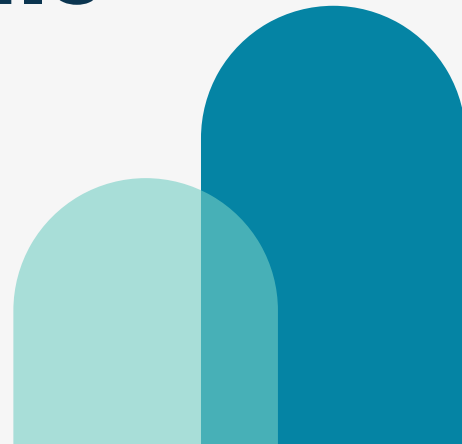




Transfer Learning Framework for Epileptic Event Classification from Low channel EEG signals

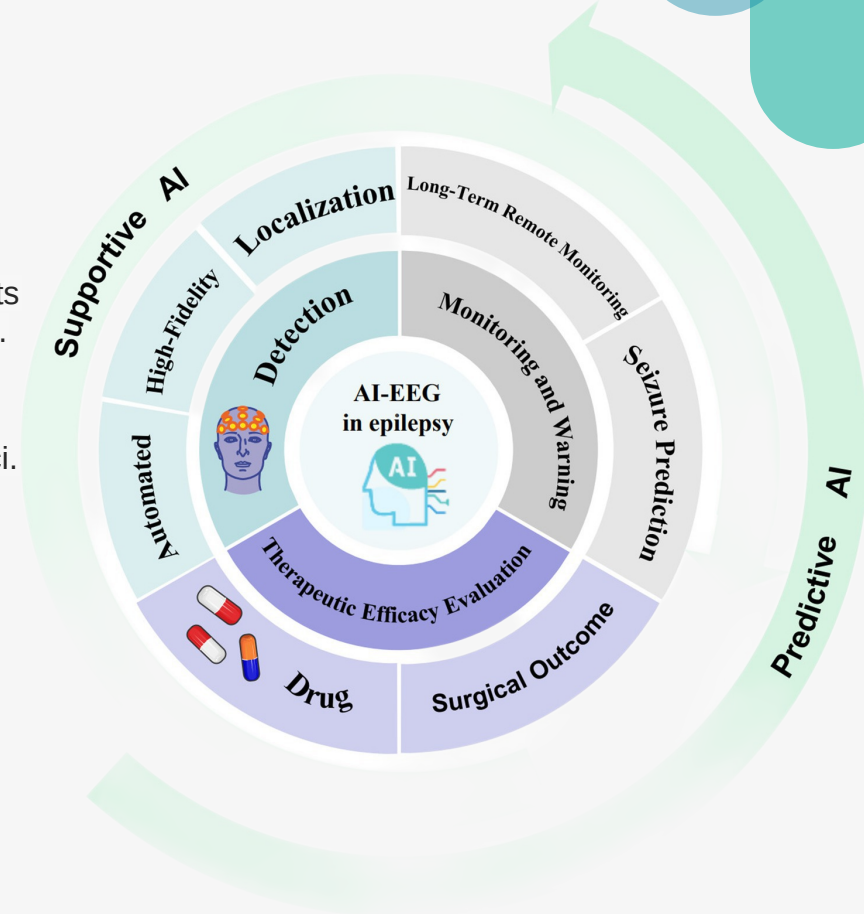
Presented by: Aditya Anand
Supervisor: Dr. Tushar Sandhan



Why Epilepsy Detection Matters?

- Epilepsy affects approximately 50 million individuals globally. The WHO estimates that nearly 70% of patients with epilepsy can achieve freedom from seizures with appropriate diagnosis and treatment.
- According to a survey in china, two-thirds of epilepsy patients fails to receive adequate treatment due to gaps in diagnosis.
- EEG (ElectroEncephaloGraphy) plays critical role in diagnosis, classification and localization of epileptogenic foci. In theory, all epileptic seizures can be associated with epileptiform discharges on the EEG
- Visual EEG classification is subjective, error prone and time consuming. Also, real time monitoring for whole day is not feasible for such number of patients.

Recent study shows, AI can be as efficient as experts at EEG seizure detection.



Project timeline:

01

Backbone Model

Used backbone 1D CNN model which works on classifying emotions via EEG signals

02

Data Mismatch fix and low signal preprocessing

Implemented a strategy to convert four channel EEG to mono channel signal

03

Baseline Model

Model collapse

04

Trying strategies to account for data imbalance

Normal:Abnormal ratio in dataset was 5.6:1

05

XAI for analysis

Tried different plots like t-SNE and Saliency plots to understand the model

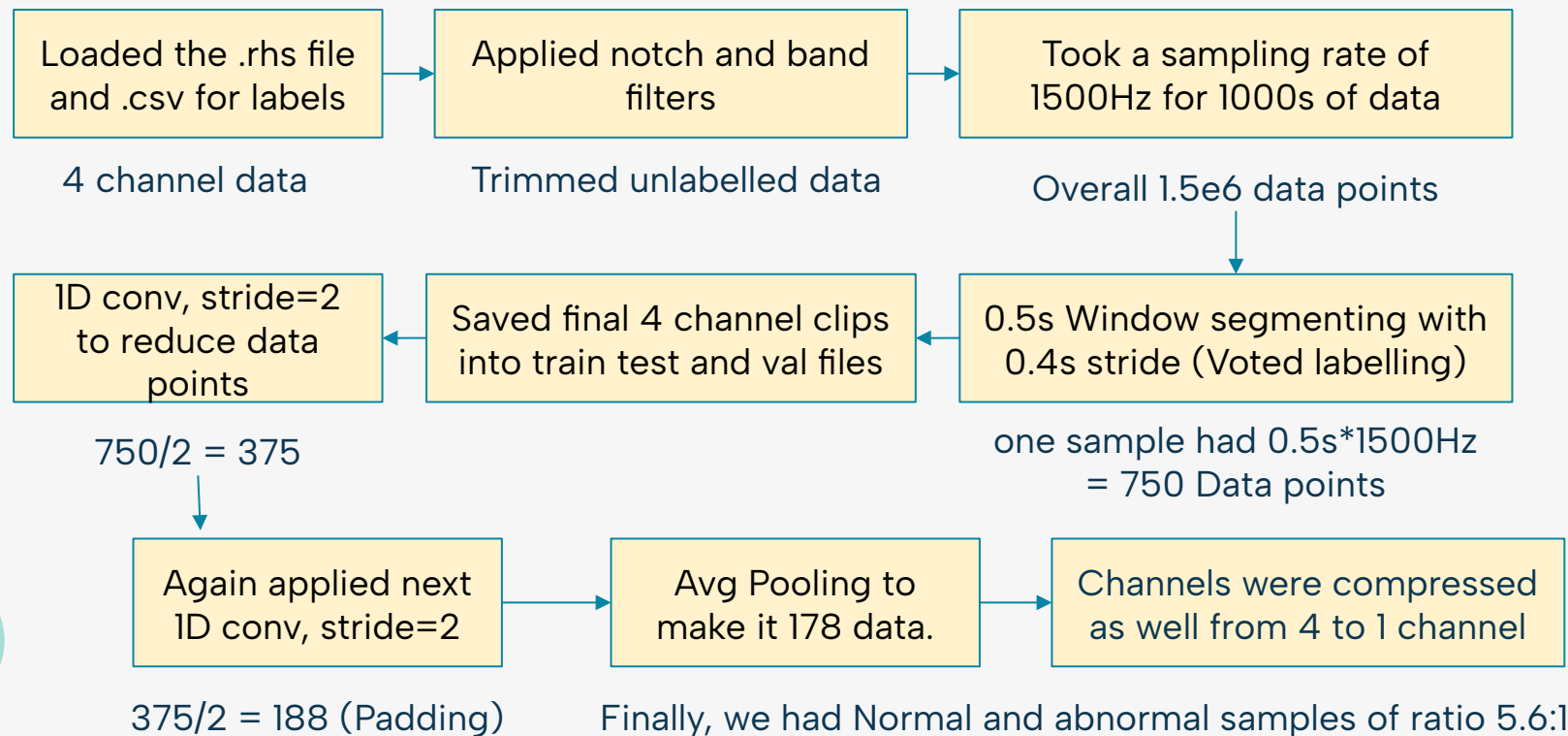
06

Shift to transformer backbone

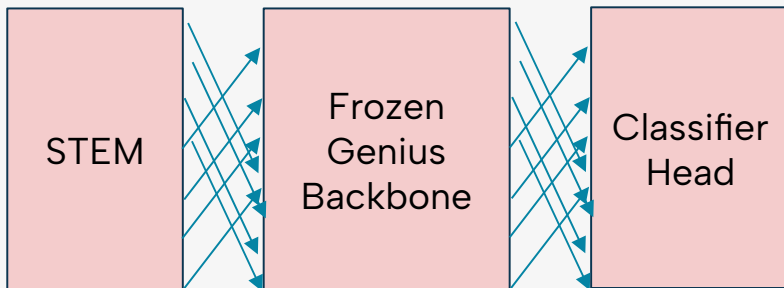
New backbone EEGFORMER

Backbone mismatch issue!

The backbone model was designed to handle mono channel and expected 178 data points!



Baseline Model



This model achieved an accuracy of ~89% , looks promising at first

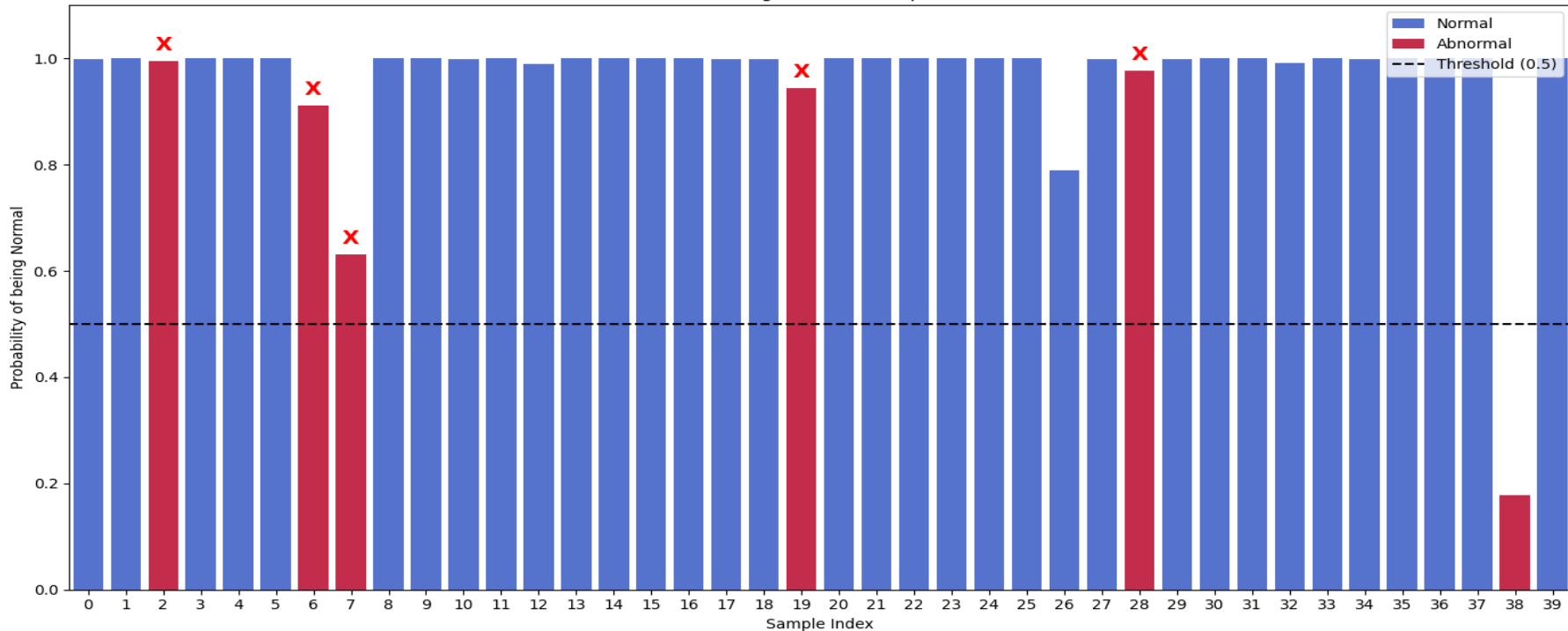
Final Eval (Best Val F1: 0.2826)

	precision	recall	f1-score	support
Normal	0.89	0.99	0.94	421
Abnormal	0.88	0.30	0.45	76
accuracy			0.89	497
macro avg	0.89	0.65	0.69	497
weighted avg	0.89	0.89	0.86	497

Notice how
5.6/6.6 ~ 85%

Baseline Model

Model Confidence (First 40 Contiguous Test Samples) - NORMAL PROBABILITY



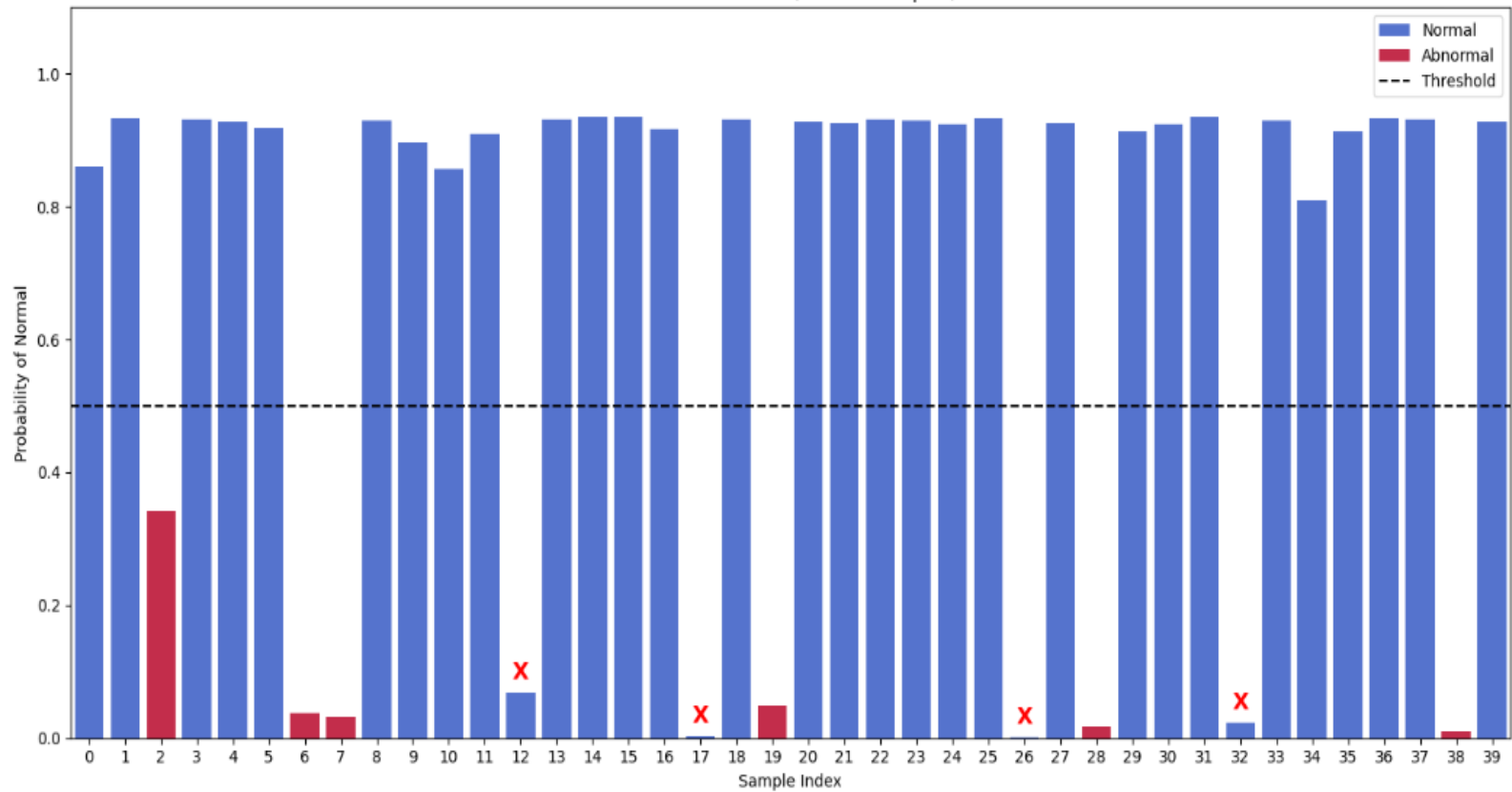
This made me remember my school days
when I had a binary quiz on history

Adding weighted random sampler:

Idea was to use weighted random sample, It gives rare samples a higher chance of being picked, so every batch looks balanced even if the dataset is not.

	precision	recall	f1-score	support
Normal	0.97	0.95	0.96	421
Abnormal	0.73	0.82	0.77	76
accuracy			0.93	497
macro avg	0.85	0.88	0.86	497
weighted avg	0.93	0.93	0.93	497
Test Results: Accuracy=92.56%, F1=0.7702, AUROC=0.9439				

Model Confidence (First 40 Samples)



SMOTE + ENN

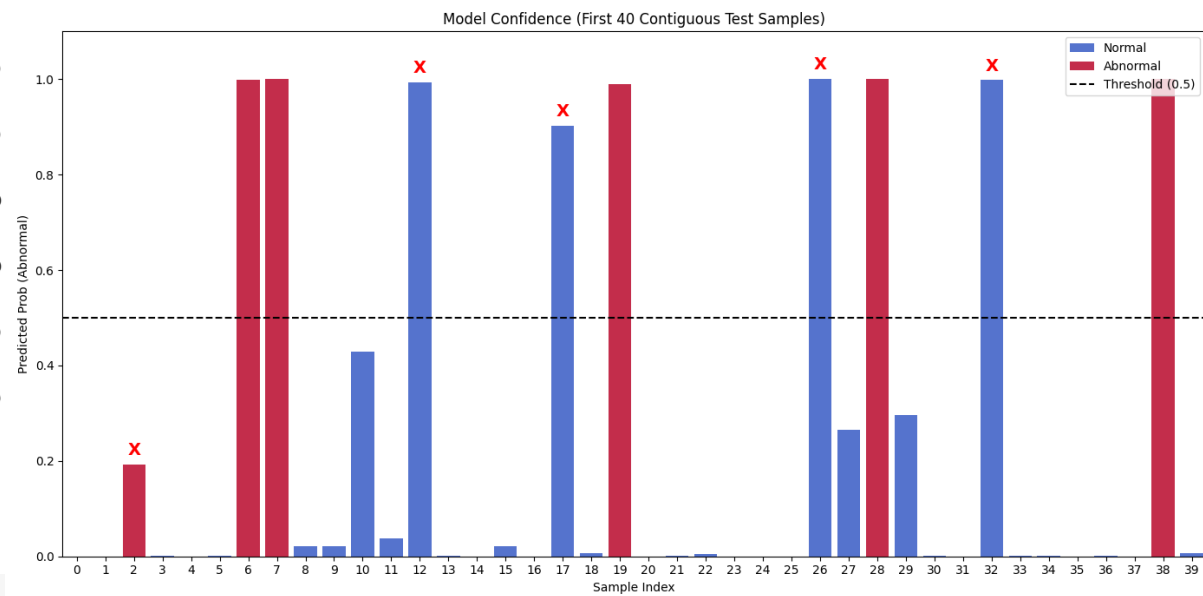
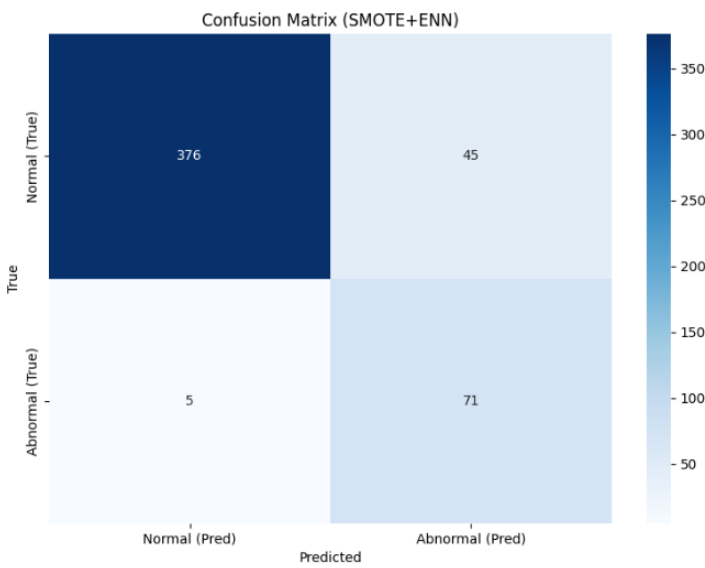
SMOTE: Creates synthetic abnormal features to increase minority class

ENN : Sharpens the boundary by removing close neighbours

Final Eval (Best Val F1: 0.7391)

	precision	recall	f1-score	support
Normal	0.99	0.89	0.94	421
Abnormal	0.61	0.93	0.74	76
accuracy			0.90	497
macro avg	0.80	0.91	0.84	497
weighted avg	0.93	0.90	0.91	497

Test AUROC: 0.9517



Recall is >90% but we lost precision

We see that many normal are now labelled as abnormal.. Which decreases the precision value, but now every abnormal is almost correctly identified (high recall value)

Is there a way, we can sort of use both our models to get a decent trade off between recall and precision?



Ensemble model with tuned threshold

Ensemble Model Report:

	precision	recall	f1-score	support
Normal	0.97	0.94	0.96	421
Abnormal	0.71	0.86	0.78	76
accuracy			0.93	497
macro avg	0.84	0.90	0.87	497
weighted avg	0.93	0.93	0.93	497

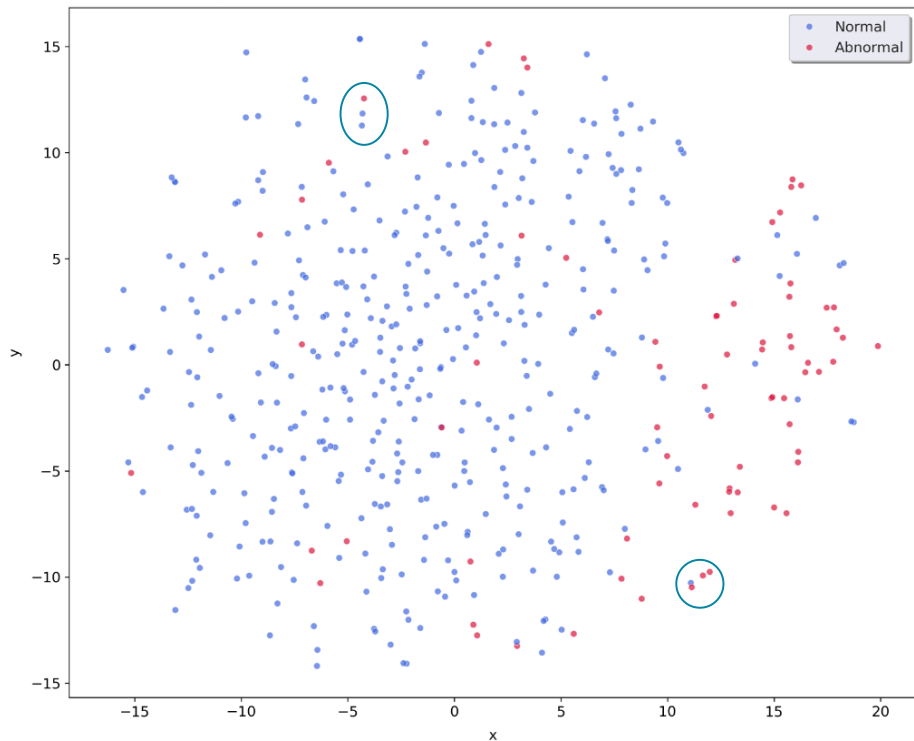
Results: Accuracy=92.56%, F1=0.7784, AUROC=0.9494

I empirically found threshold of 0.7 on probability gave best outcome!

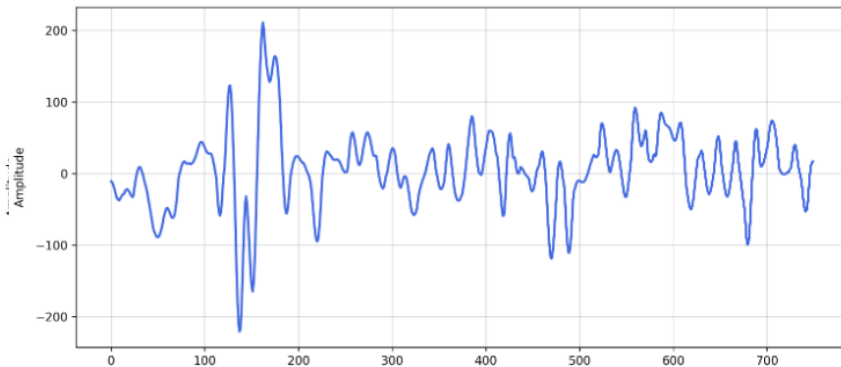
Explain the results we got:

t-SNE plot analysis:

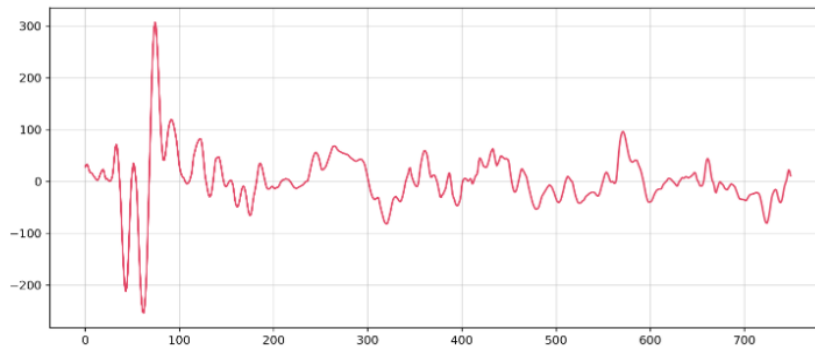
Stable Model (F1=0.70): Clearly Separated Clusters



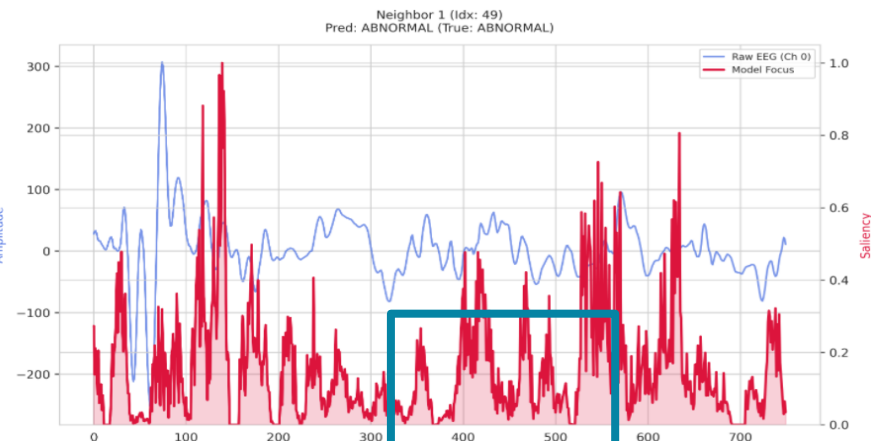
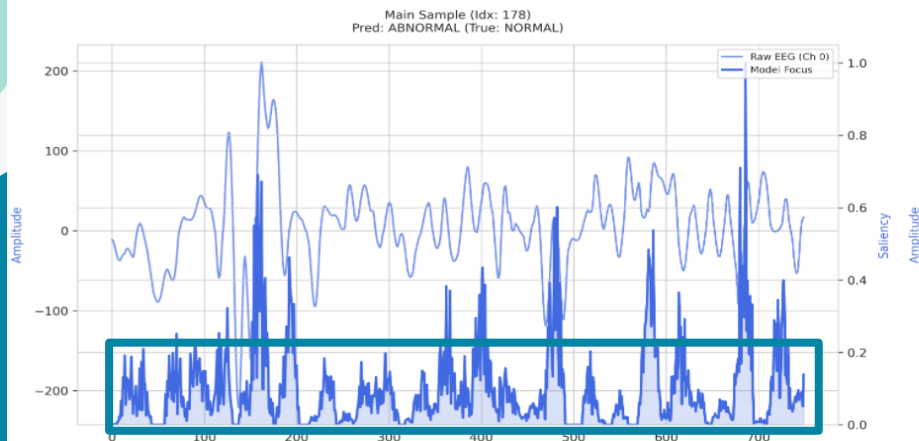
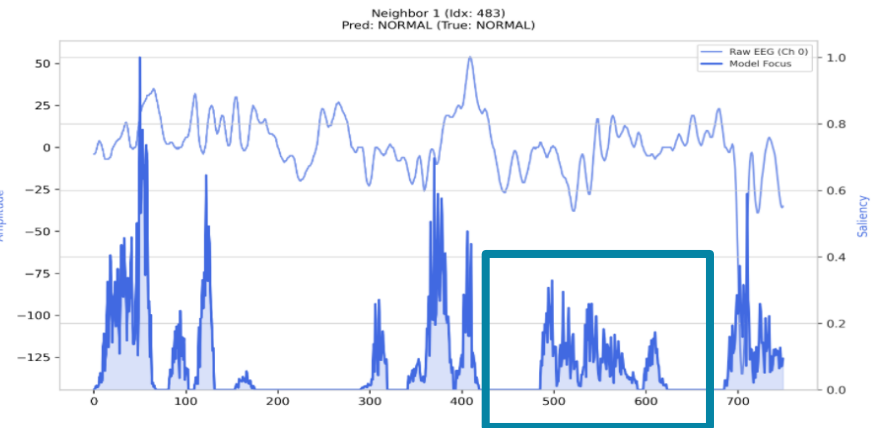
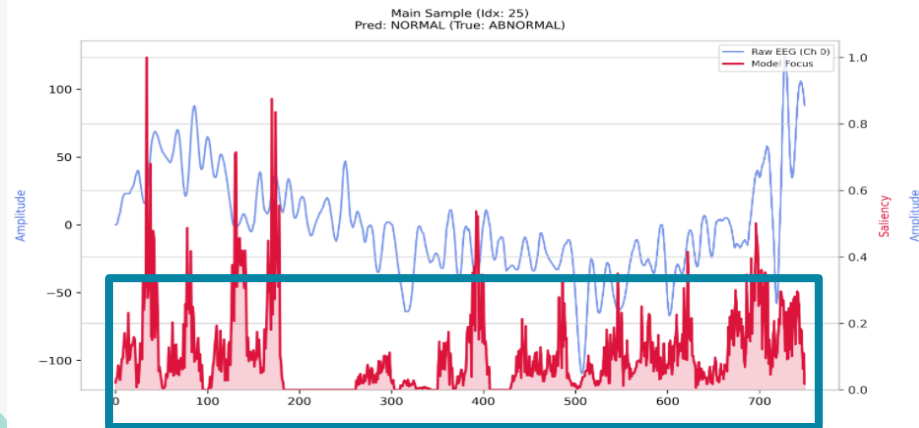
Main Sample (Index: 178) | TRUE LABEL: NORMAL



Neighbor 1 (Index: 49) | TRUE LABEL: ABNORMAL



Saliency plots:



Replace backbone model with EEG FORMER

Using CNNs for time series data led to a dead end...

They can recognise low level features/patterns of features using filters like A "sharp spike" filter. A "slow wave" filter. A "10 Hz oscillation" filter.

- Now, for backbone we tried using EEG Former model available online (we tried to move from isolated windows to sequence models)
- Unfortunately, it wasn't pretrained so I had to train it on kaggle dataset inside PIL lab systems for 30 epochs (~97% accuracy) as the Kaggle task was easy (telling a full seizure from a normal signal)

Best Model from Epoch: 29

Accuracy: 0.8934

Precision: 0.6018

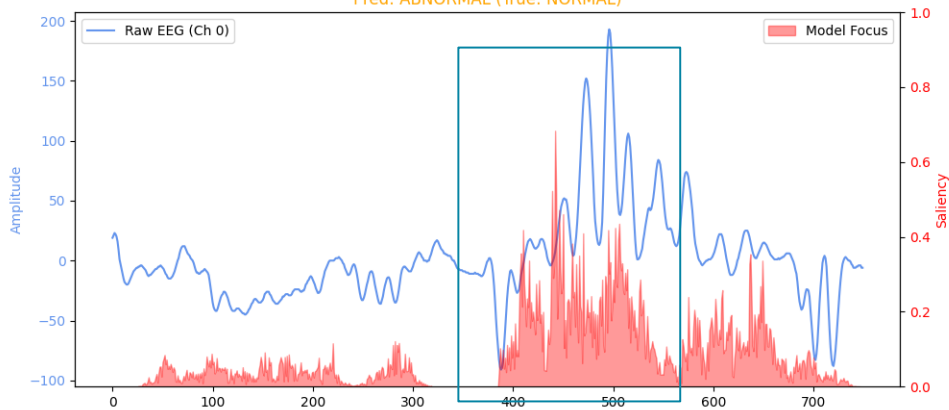
Recall: 0.8947

F1 Score: 0.7196

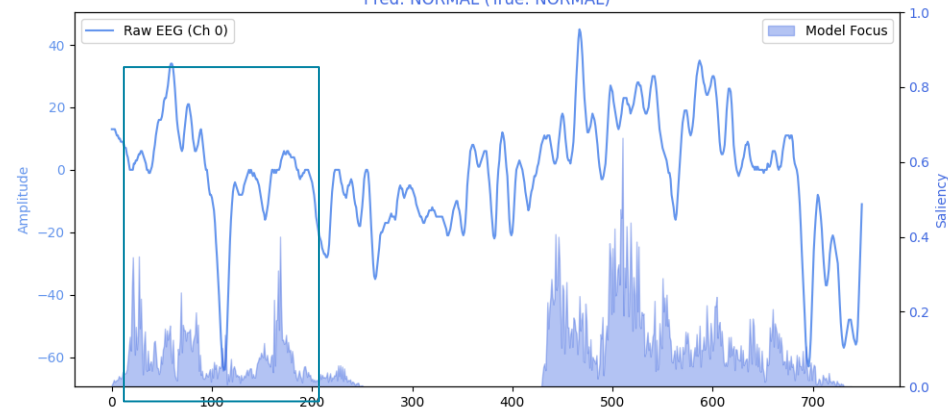
AUROC: 0.9347

- This is the result of our baseline model after limited training and dataset.
- Transformers tend to look at whole signal as once instead of looking at regional features.

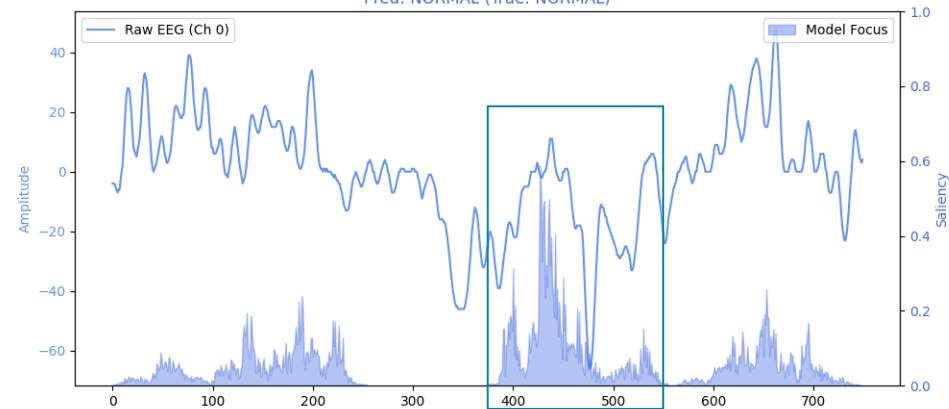
Sample (Idx: 38)
Pred: ABNORMAL (True: NORMAL)



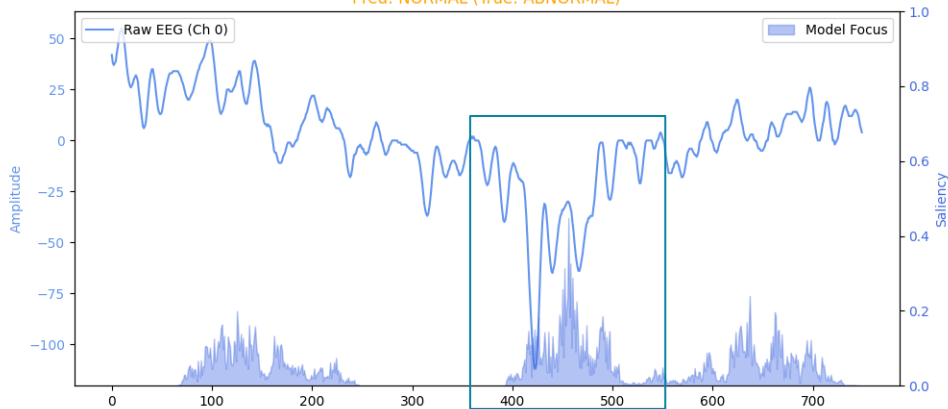
Sample (Idx: 103)
Pred: NORMAL (True: NORMAL)



Sample (Idx: 489)
Pred: NORMAL (True: NORMAL)



Sample (Idx: 88)
Pred: NORMAL (True: ABNORMAL)





**Thank
you!**





Appendix

Failed tries!

Pretraining the head

Worse results, Pretraining the head on garbage features simply trains it to memorize noise.

Semi Supervised Learning

Reached AUROC of 0.18, model became stupid genius. Unlabelled dataset was filled with artifacts and noise!

Use Focal Loss

Model became very unstable. Loss and F1 score curve were horrible

Data balancing

No improvement at all!

Limitations!

**Uneven dataset (5.6:1 ratio) Individual dataset, single mice for 2000s
(Also included artifacts)**

**We lost information of high resolution when we compressed the
dataset to obtain 177 datapoints**

**Once trained, the model is fixed. It does not adapt to a new
patient's baseline EEG or learn from corrections made by doctors
over time.**

**The backbone model in EEG former required rigorous training on
large datasets**

Future Scopes

Cross Subject validation

- If it can maintain its F1 score on different subject, that will be the best test for model quality

Advance Data Augmentation

- Apart from SMOTE + ENN we can try doing noise injection, time warping, amplitude scaling to make model more robust

Native Architecture

- Instead of transfer learning, we can make a whole transformer model and train it on varying subject data in large amount

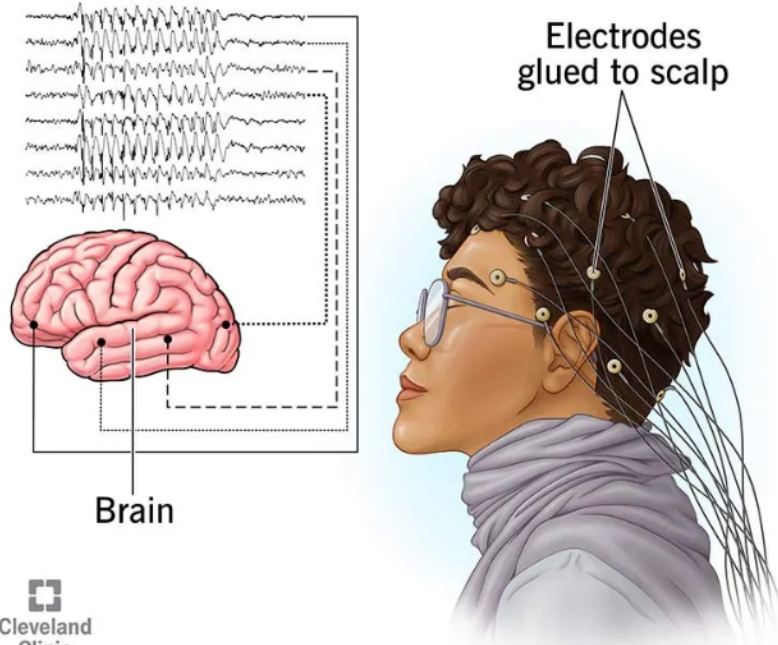
Channel data use

- Instead of compressing the channels, we can try to use spatial convolutions or use all channel in training altogether

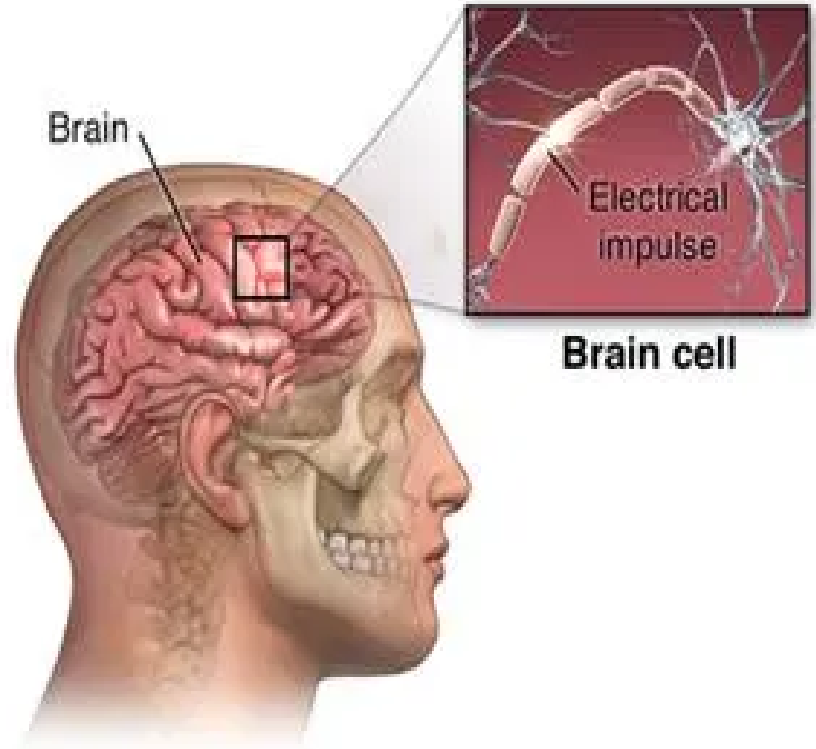
EEG signals?

Electroencephalogram (EEG)

EEG (scan of brainwaves)



Electrical brain activity



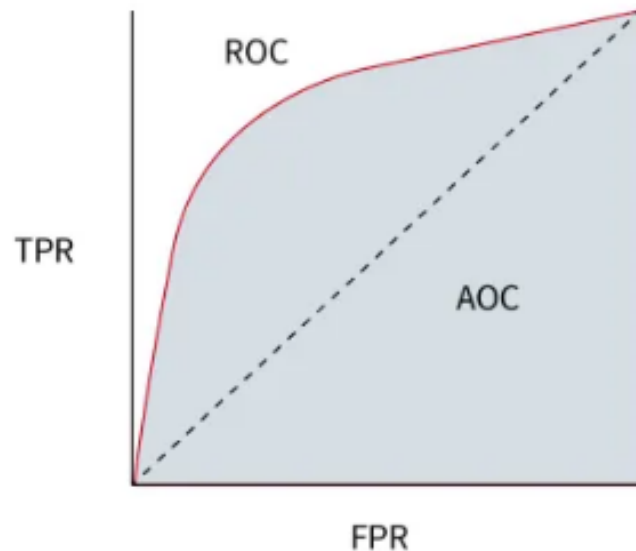
Performance Metrics:

Python

		Predicted	
		+ve	-ve
Expected	+ve	TP	FN
	-ve	FP	TN

$$TPR = \frac{TP}{TP + FN} \quad FPR = \frac{FP}{FP + TN}$$

$$Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

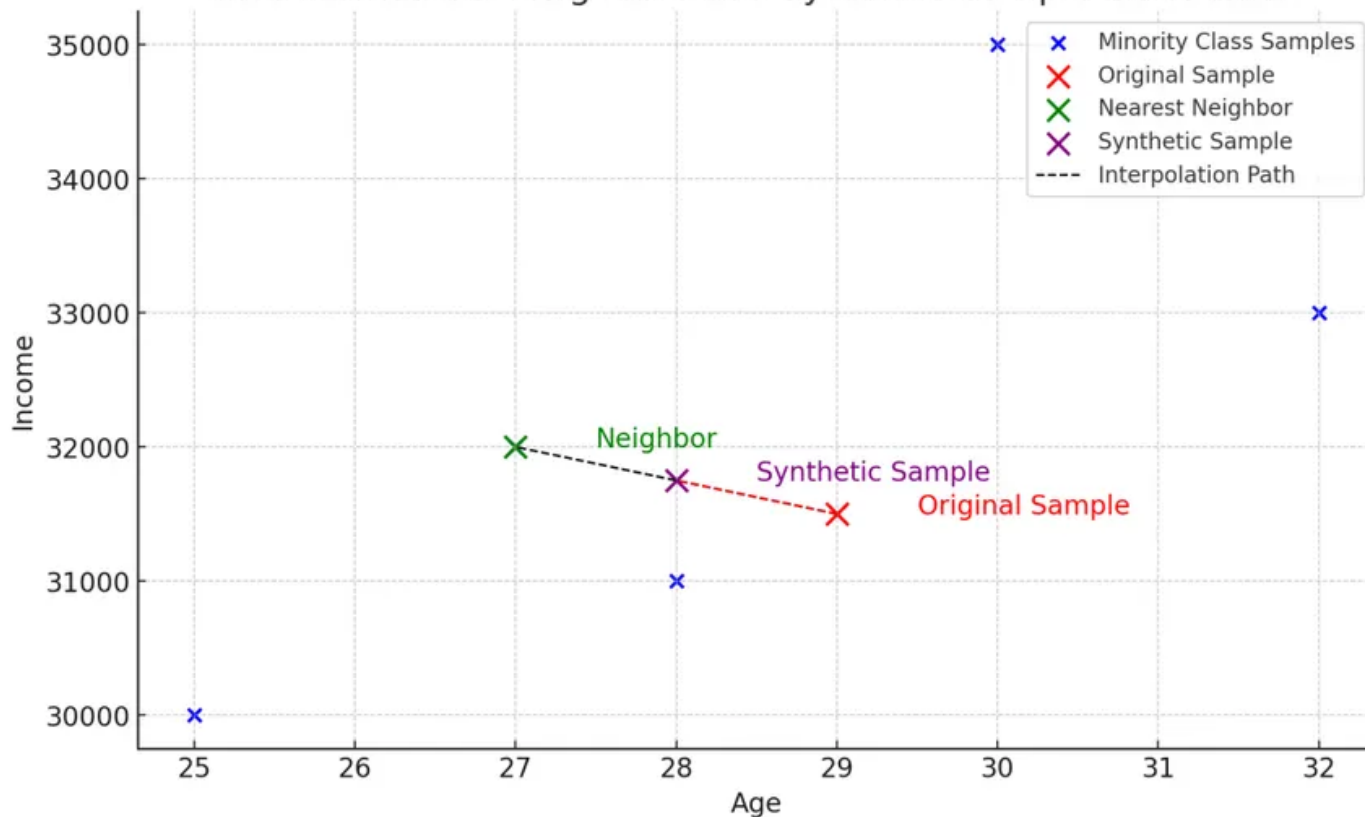


$$AUROC = P(\hat{y}_{pos} > \hat{y}_{neg})$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

Methods explained:

SMOTE: Nearest Neighbors and Synthetic Sample Generation



SMOTE: Synthetic Minority Oversampling Technique

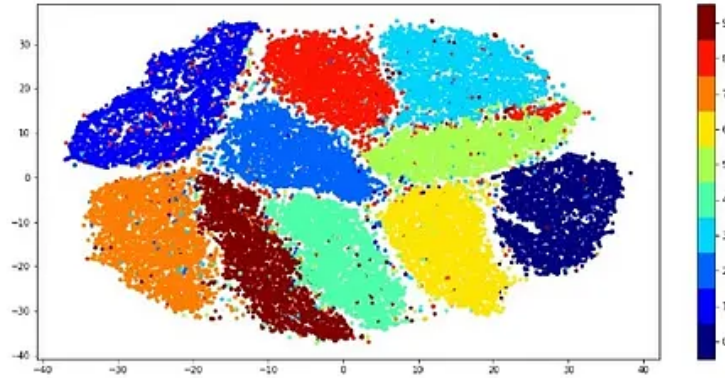
It generates more minority data using basic interpolation synthetically

$$x_{new} = x_i + \lambda(x_{near} - x_i)$$

ENN: Edited Nearest Neighbors
Removes borderline unstable points

t-SNE = t-Distributed Stochastic Neighbor Embedding

MNIST - TSNE



Compute pairwise distances



Convert distances → high-dim probabilities

$$p_{j|i} = \frac{\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{\|x_i - x_k\|^2}{2\sigma_i^2}\right)}$$



Convert to low-dim student t-probability

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$



Compute KL loss

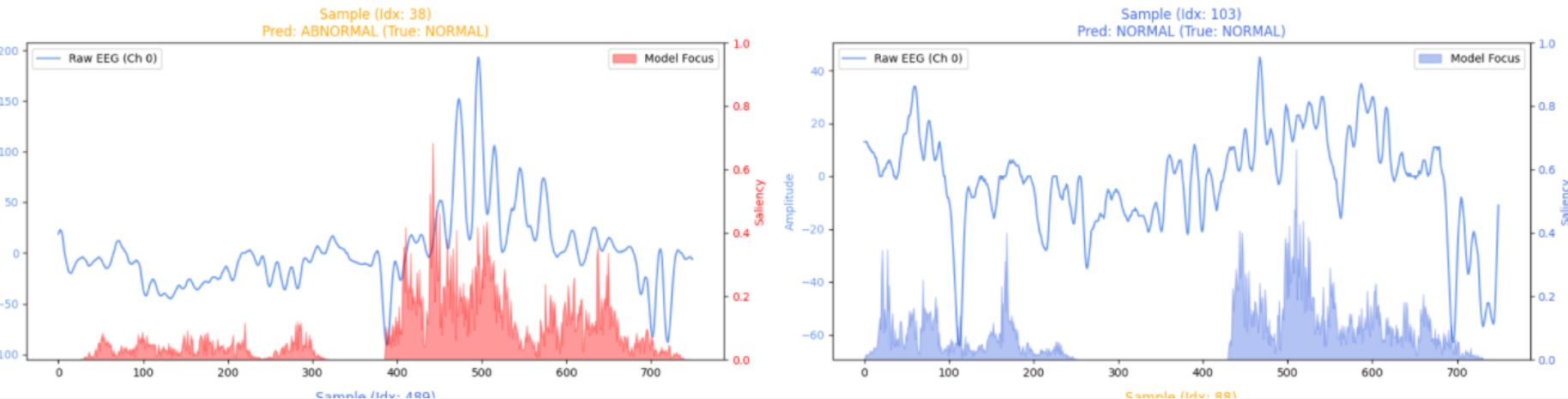
$$\frac{\partial KL}{\partial y_i} = 4 \sum_j (p_{ij} - q_{ij})(y_i - y_j)(1 + \|y_i - y_j\|^2)^{-1}$$

Optimize Y using gradient descent

$$KL(P \parallel Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$



Saliency maps: Understanding What the Model Focuses On



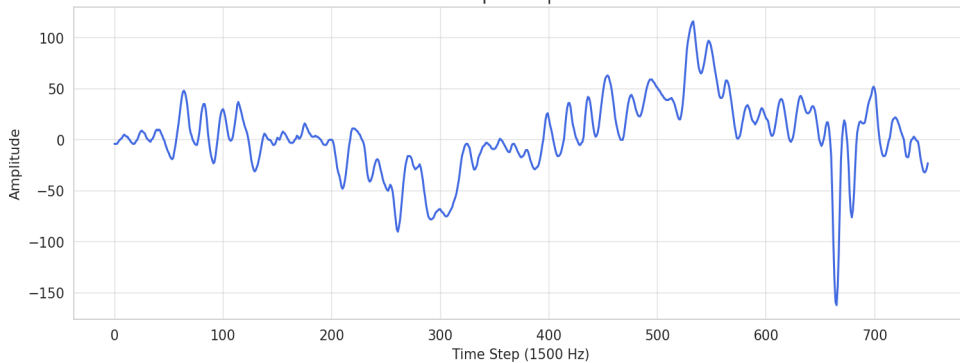
Forward pass to get the class score (logit):
 $y=f(x, \dots)$ say, x is data point in a sample

$$S = \left| \frac{\partial y_{class}}{\partial x} \right|$$

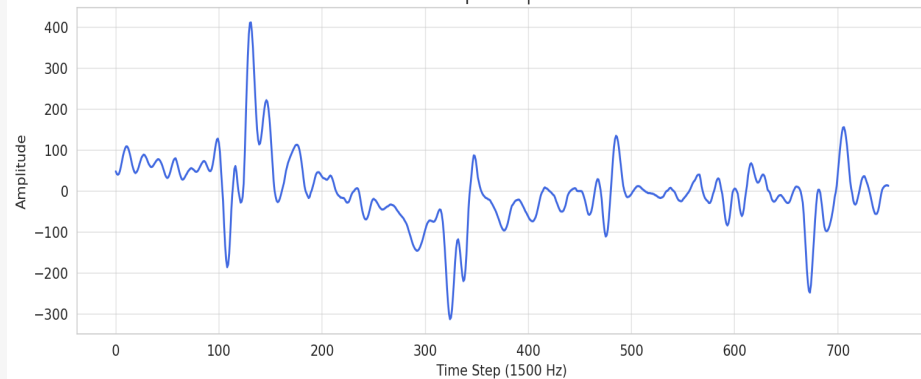
Normalize and plot the saliency:
brighter = model strongly depends on
this EEG region
darker = less important

Samples 12,17,26,32:

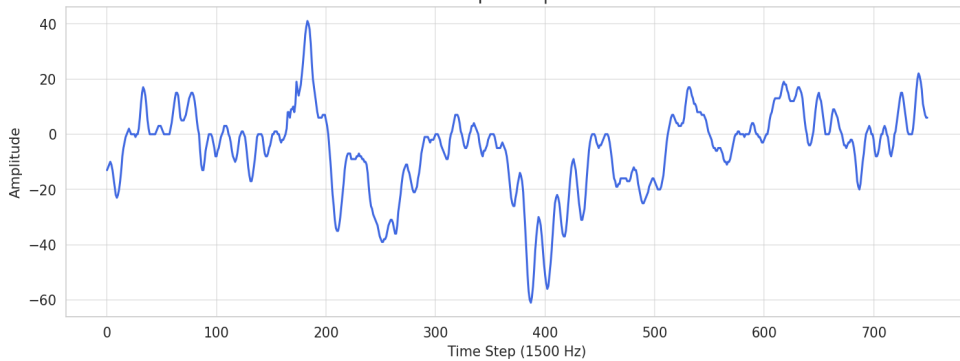
Waveform for Sample 12 | True Label: Normal



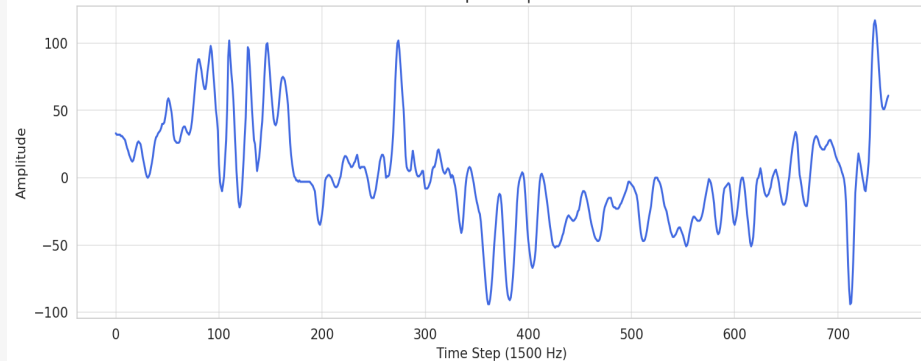
Waveform for Sample 26 | True Label: Normal



Waveform for Sample 17 | True Label: Normal

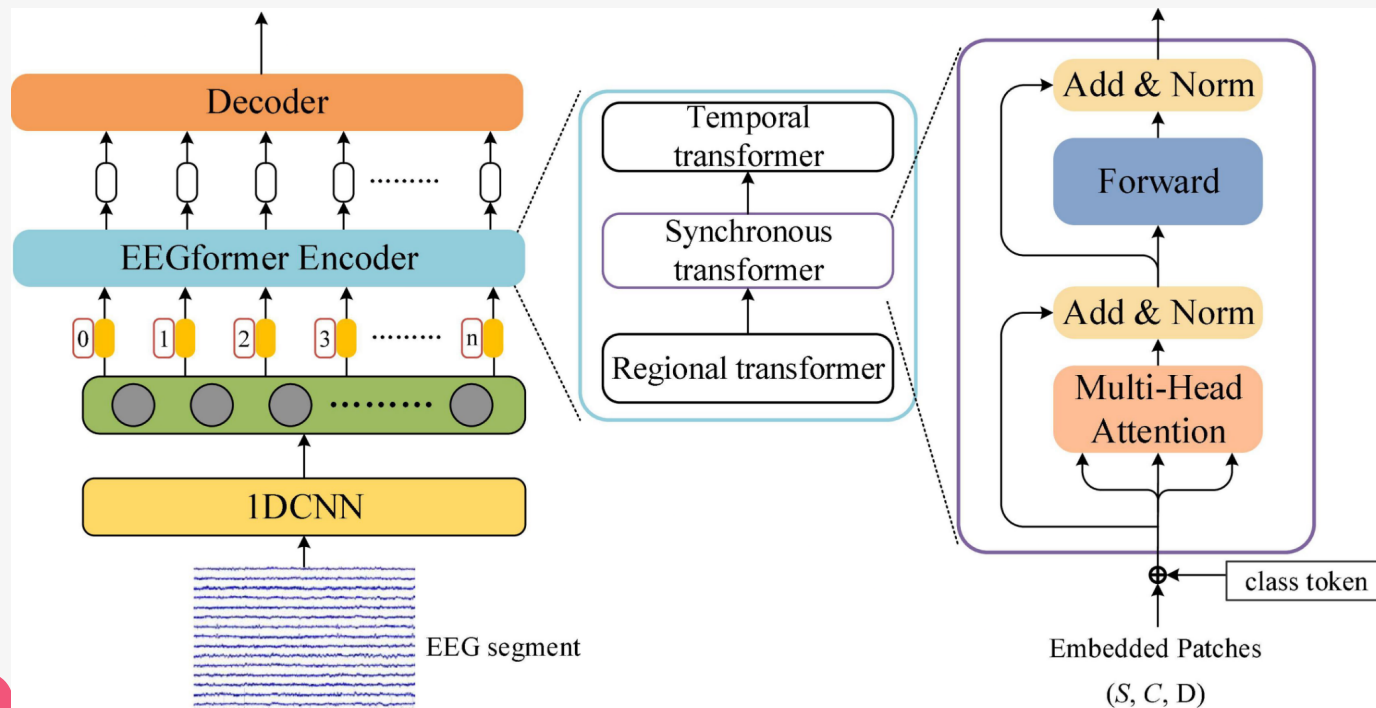


Waveform for Sample 32 | True Label: Normal



EEG Former:

Transformer based approach!



Credits:

- ZHIZIANG WAN (2023). *Frontiers in Neuroscience*
 - EEGformer: A transformer-based brain activity classification method using EEG signal
- FERIR. (2024). *Github repo of unofficial EEGformer model*
 - Unofficial implementation of EEG Former paper
- EMADELDEEN24. (2021). *Github Repo of 1D pretrained CNN*
 - Link to repository
- DONG ZHOU. (2021). *Data on epilepsy affected population (1st slide)*
 - Epilepsy in China: major progress in the past two decades
- VAN DER MAATEN & HINTON, 2008. (YEAR). *t-SNE plot algorithm*
 - Visualise data using t-SNE
- KAREN SIMONYAN (2014), Saliency map representation idea.
 - Classification Models and Saliency Maps